

International Cybersecurity and Digital Forensics Academy (ICDFA)



SBT-DF203 Lab 6 Report

Firewall Traffic Control and Forensic Verification

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Student ID:	2025-FWSD-11392
Course Title:	Basic Networking Skills for Digital Forensics
Course Code:	SBT-DF203
Lab Number:	Lab 6
Lab Title:	Firewall Traffic Control and Forensic Verification
Instructor:	Aminu Idris, AMCPN
Case/Lab Identifier:	SBT-DF203-Lab6-Abubakari-Sadique Hamidu
Date:	18 September 2026

1. Source Materials and Slide Alignment

This practical aligns directly with the SBT-DF203 lecture series covering Network-Based and

Host-Based Firewalls, Linux Netfilter evaluation chains (INPUT, OUTPUT, FORWARD), iptables rule matching criteria, and packet interception concepts. The practical verifies HTTP access on Apache, exports and hashes baseline rulesets, inserts a narrow DROP rule, captures blocked retransmissions, correlates rule hit counters, and verifies clean restoration.

2. Executive Summary

A host-based firewall verification was conducted inside an isolated Kali Linux Docker container (sbtdf203-lab) running Apache2. The objective was to capture a baseline allowed HTTP session, export and hash the existing ruleset, insert a source-specific DROP rule targeting TCP port 80 at position 1 of the INPUT chain, capture the resulting connection timeout and retransmissions, evaluate iptables packet/byte hit counters, and restore original access.

3. Learning Outcomes Demonstrated

- Differentiated host-based and network-based firewall deployment models.
- Interpreted Linux Netfilter INPUT, OUTPUT, and FORWARD evaluation chains.
- Exported, hashed, verified, and removed iptables rule specifications safely.
- Captured and analyzed allowed (200 OK) and blocked (Connection timed out) HTTP sessions.
- Explained the forensic differences between DROP (silent packet discard) and REJECT (active TCP RST or ICMP error response).

4. Legal, Ethical and Safety Rules Compliance

- Executed analysis strictly within an authorized, isolated lab container (sbtdf203-lab).
- Exported and hashed the original ruleset (iptables_before.rules) before applying any changes.
- Inserted only a narrowly scoped source IP rule on TCP port 80 without flushing global

policy.

- Removed the rule immediately after capturing evidence and verified access restoration.

5. Tools and Environment Used

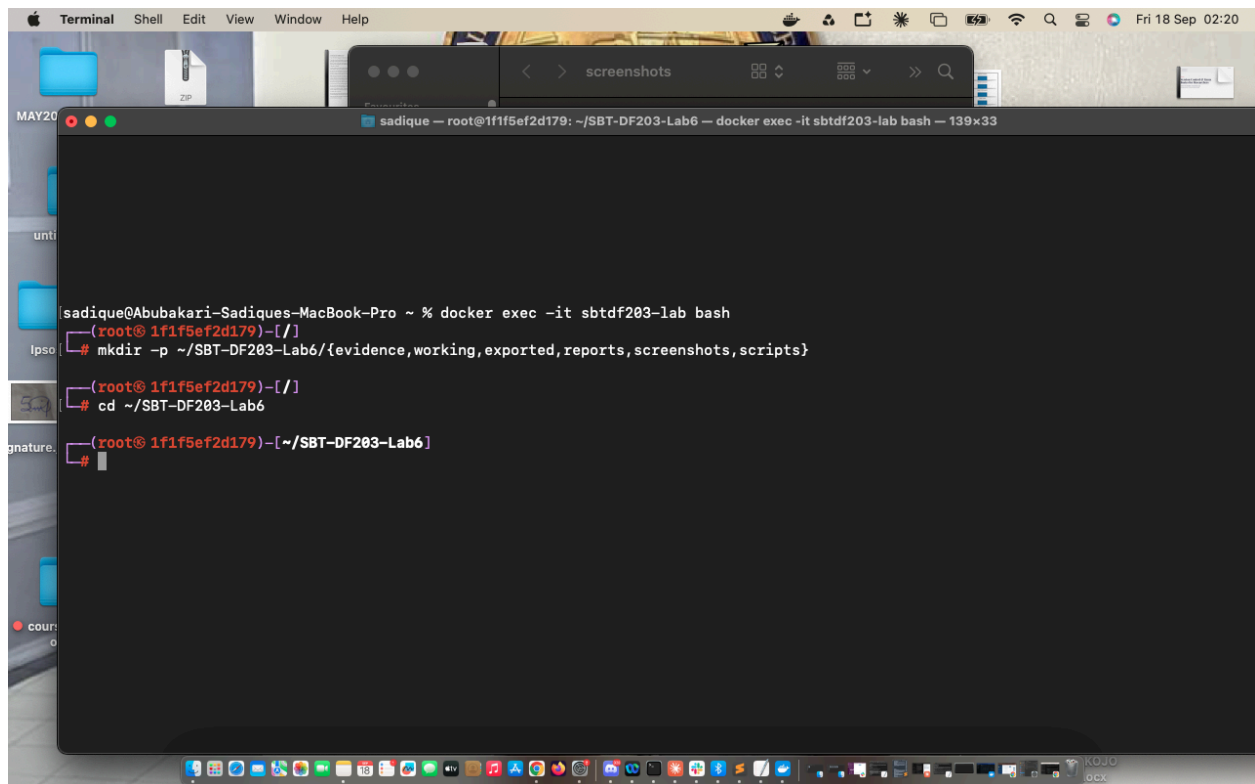
Item	Tool Used	Forensic Purpose
Operating System	Kali Linux (kali-rolling container)	Workstation hosting Apache and Netfilter.
Web Server	Apache2 (v2.4.68)	Local HTTP service hosting firewall_lab.html.
Firewall Utility	iptables & iptables-save	Managing and exporting Netfilter rule chains.
Packet Capture	TShark & Wireshark (v4.6.6)	Capturing allowed and blocked TCP port 80 frames.
Client Testing	curl	Testing HTTP connection outcomes and timeouts.
Hashing Utility	sha256sum	Evidence chain-of-custody checksums.

6. Lab Folder Structure and Evidence Preparation

The standard forensic workspace hierarchy was established before analysis:

SBT-DF203-Lab6/

- |— evidence/ # Preserved original capture files (http_allowed.pcapng, http_blocked.pcapng)
- |— working/ # Working copies for active investigation
- |— exported/ # Exported objects and payloads
- |— reports/ # TSV extractions, hashes, and text summaries
- |— screenshots/ # Contemporaneous terminal evidence
- |— scripts/ # Automation scripts



The screenshot shows a macOS desktop environment. In the foreground, a terminal window is open, displaying a series of commands and their outputs. The terminal prompt is `sadique@Abubakari-Sadiques-MacBook-Pro ~`. The user has entered `% docker exec -it sbtdf203-lab bash`, which has opened a root shell inside a container. The prompt changes to `(root@1f1f5ef2d179)-[/]`. The user then enters `# mkdir -p ~/SBT-DF203-Lab6/{evidence,working,exported,reports,screenshots,scripts}`, creating the directory structure. The prompt returns to `(root@1f1f5ef2d179)-[/]`. Finally, the user enters `# cd ~/SBT-DF203-Lab6`, changing the directory. The prompt now shows `(root@1f1f5ef2d179)-[~/SBT-DF203-Lab6]`. In the background, a file explorer window is open, showing a folder named `screenshots`. The desktop also shows a dock with various application icons at the bottom and a status bar at the top right indicating the date and time as `Fri 18 Sep 02:20`.

```
sadique@Abubakari-Sadiques-MacBook-Pro ~ % docker exec -it sbtdf203-lab bash
(root@1f1f5ef2d179)-[/]
# mkdir -p ~/SBT-DF203-Lab6/{evidence,working,exported,reports,screenshots,scripts}
(root@1f1f5ef2d179)-[/]
# cd ~/SBT-DF203-Lab6
(root@1f1f5ef2d179)-[~/SBT-DF203-Lab6]
```

```
Terminal Shell Edit View Window Help
sadique — root@1f1f5ef2d179: ~/SBT-DF203-Lab6 — docker exec -it sbtdf203-lab bash — 139x33

(root@1f1f5ef2d179)~/~/SBT-DF203-Lab6
# apt update && apt install -y apache2 curl iptables tshark wireshark net-tools
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Err:1 http://kali.download/kali kali-rolling InRelease
Sub-process /usr/bin/sqv returned an error code (1), error message is: Verifying signature: Not live until 2026-09-22T00:09:24Z
Fetched 34.0 kB in 2s (20.3 kB/s)
60 packages can be upgraded. Run 'apt list --upgradable' to see them.
Warning: An error occurred during the signature verification. The repository is not updated and the previous index files will be used. Signature verification failed: http://kali.download/kali kali-rolling InRelease: Sub-process /usr/bin/sqv returned an error code (1), error message is: Verifying signature: Not live until 2026-09-22T00:09:24Z
Warning: Failed to fetch http://http.kali.org/kali/dists/kali-rolling/InRelease Sub-process /usr/bin/sqv returned an error code (1), error message is: Verifying signature: Not live until 2026-09-22T00:09:24Z
Warning: Some index files failed to download. They have been ignored, or old ones used instead.
curl is already the newest version (8.21.0-2).
tshark is already the newest version (4.6.6-1).
wireshark is already the newest version (4.6.6-1).
net-tools is already the newest version (2.10-2).
Upgrading:
  apache2  apache2-bin  apache2-data  apache2-utils
Installing:
  iptables
Installing dependencies:
  libip4tc2  libip6tc2  libnetfilter-conntrack3  libnfnetlink0  libnftables1  libnftnl11  nftables
Suggested packages:
  firewallld  kmod
Summary:
  Upgrading: 4, Installing: 8, Removing: 0, Not Installed: 54
```

```
Terminal Shell Edit View Window Help
sadique — root@1f1f5ef2d179: ~/SBT-DF203-Lab6 — docker exec -it sbtdf203-lab bash — 132x33

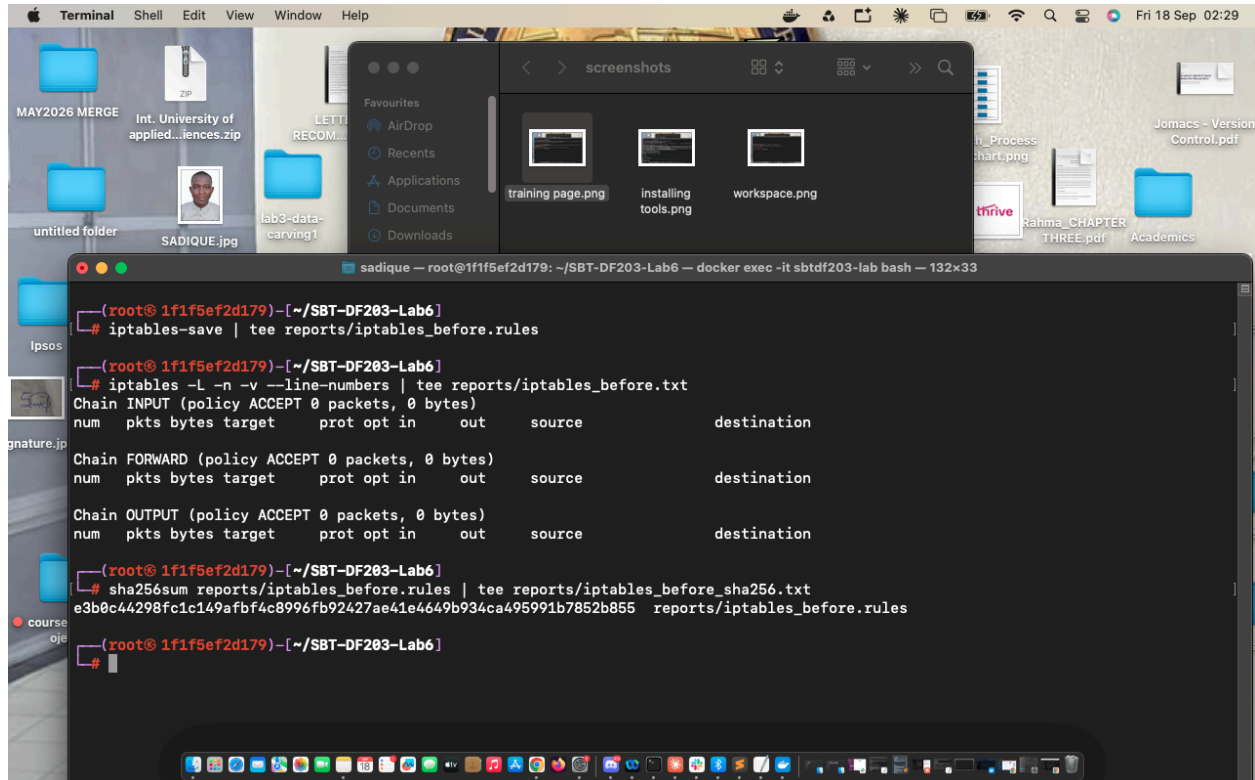
(root@1f1f5ef2d179)~/~/SBT-DF203-Lab6
# service apache2 start
Starting Apache httpd web server: apache2AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 172.17.0.2. Set the 'ServerName' directive globally to suppress this message
.

(root@1f1f5ef2d179)~/~/SBT-DF203-Lab6
# printf '<!DOCTYPE html>\n<html><body><h1>ICDFA Network Forensics Firewall Lab</h1><p>Server: sbtdf203-lab</p></body></html>\n' | tee /var/www/html/firewall_lab.html
<!DOCTYPE html>
<html><body><h1>ICDFA Network Forensics Firewall Lab</h1><p>Server: sbtdf203-lab</p></body></html>

(root@1f1f5ef2d179)~/~/SBT-DF203-Lab6
#
```

Original rulesets and packet captures were hashed immediately upon generation:

- reports/iptables_before.rules (Hashed in iptables_before_sha256.txt)
- evidence/http_allowed.pcapng SHA-256:
3377a18773be56742321a6e62bab421a17d89053e3b7a688dc5cdcdcec1e37f3
- evidence/http_blocked.pcapng SHA-256:
59427b620b48565bdff06dacf1d11015c66204e69523cf974d257aab6d8f4be3



7. Evidence and Chain-of-Custody Worksheet

Field	Student / Forensic Entry
Case/Lab Identifier	SBT-DF203-Lab6-AbubakariSadiqueHamidu
Trainee Name	Abubakari-Sadique Hamidu

Field	Student / Forensic Entry
Date Started	2026-09-18 02:25 WAT
Evidence File(s)	http_allowed.pcapng / http_blocked.pcapng / iptables_before.rules
Source	Generated during authorized live firewall evaluation
Analysis Workstation	macOS Host / Kali Linux Docker Container (sbtdf203-lab)
Notes on Changes	Narrow DROP rule added to position 1 and completely removed after test.

8. Part A & B – Baseline HTTP Access & Allowed Capture

Baseline testing confirmed Apache listening on 0.0.0.0:80. Executing curl -v http://127.0.0.1/firewall_lab.html completed the TCP three-way handshake (SYN → SYN-ACK → ACK) and returned HTTP/1.1 200 OK.

```
Terminal Shell Edit View Window Help
sadique -- root@1f1f5ef2d179: ~/SBT-DF203-Lab6 -- docker exec -it sbtdf203-lab bash -- 132x33

(root@1f1f5ef2d179)~/SBT-DF203-Lab6
# tshark -i eth0 -f 'tcp port 80' -a duration:15 -w evidence/http_allowed.pcapng &
[1] 346

(root@1f1f5ef2d179)~/SBT-DF203-Lab6
# Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'
# sleep 2
[1]+ Done tshark -i eth0 -f 'tcp port 80' -a duration:15 -w evidence/http_allowed.pcapng

(root@1f1f5ef2d179)~/SBT-DF203-Lab6
# curl -v --connect-timeout 5 http://127.0.0.1/firewall_lab.html 2>&1 | tee reports/baseline_curl.txt
* Trying 127.0.0.1:80...
* Established connection to 127.0.0.1 (127.0.0.1 port 80) from 127.0.0.1 port 47414
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0* using HTTP/1.x
> GET /firewall_lab.html HTTP/1.1
> Host: 127.0.0.1
> User-Agent: curl/8.21.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Fri, 18 Sep 2026 03:00:21 GMT
< Server: Apache/2.4.68 (Debian)
< Last-Modified: Fri, 18 Sep 2026 02:25:15 GMT
< ETag: "73-65bb89c5b81b0"
< Accept-Ranges: bytes
< Content-Length: 115
< Vary: Accept-Encoding
< Content-Type: text/html
```

```
Terminal Shell Edit View Window Help
sadique -- root@1f1f5ef2d179: ~/SBT-DF203-Lab6 -- docker exec -it sbtdf203-lab bash -- 132x33

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0* using HTTP/1.x
> GET /firewall_lab.html HTTP/1.1
> Host: 127.0.0.1
> User-Agent: curl/8.21.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Fri, 18 Sep 2026 03:00:21 GMT
< Server: Apache/2.4.68 (Debian)
< Last-Modified: Fri, 18 Sep 2026 02:25:15 GMT
< ETag: "73-65bb89c5b81b0"
< Accept-Ranges: bytes
< Content-Length: 115
< Vary: Accept-Encoding
< Content-Type: text/html
{ [115 bytes data]
100 115 100 115 0 0 11787 0 0
* Connection #0 to host 127.0.0.1:80 left intact
<!DOCTYPE html>
<html><body><h1>ICDFA Network Forensics Firewall Lab</h1><p>Server: sbtdf203-lab</p></body></html>

(root@1f1f5ef2d179)~/SBT-DF203-Lab6
# wait

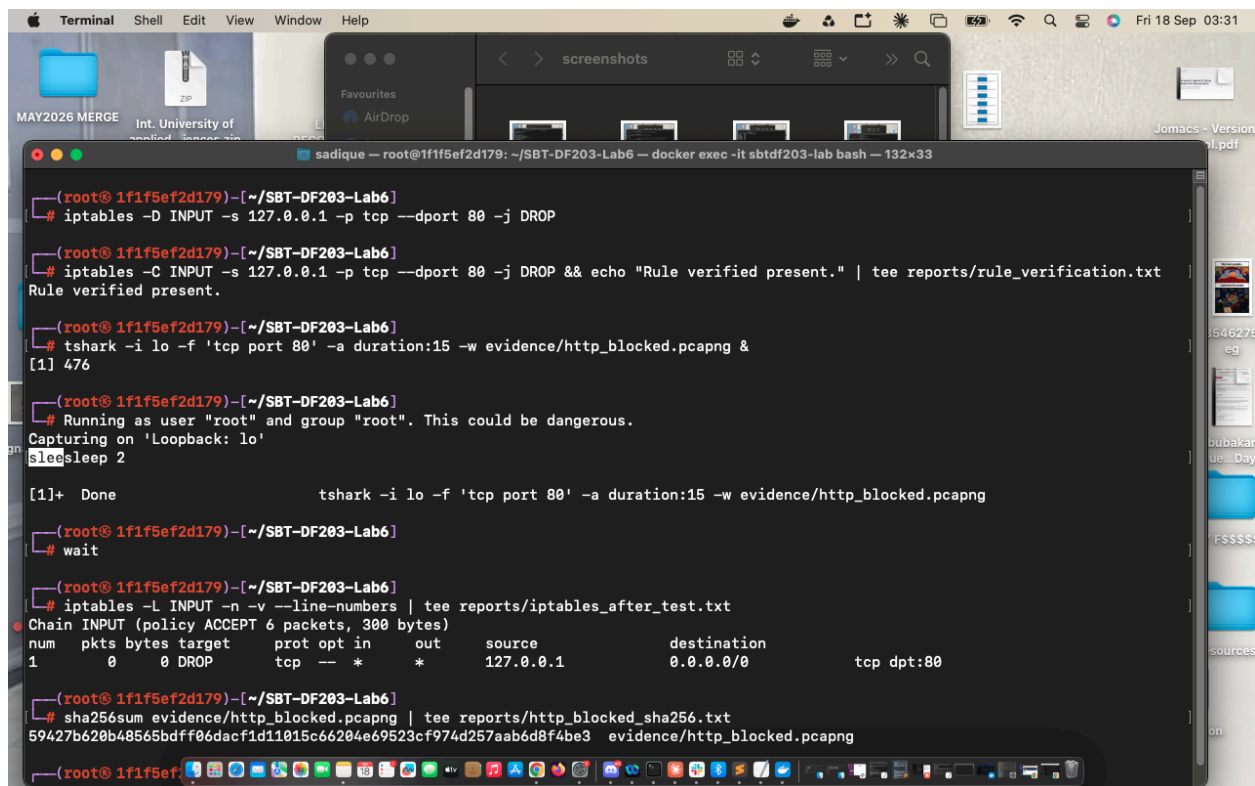
(root@1f1f5ef2d179)~/SBT-DF203-Lab6
# sha256sum evidence/http_allowed.pcapng | tee reports/http_allowed_sha256.txt
1777be1792b4c804782734ea594b36b22b5f6fdc2006ffaec1f885dc17f190b evidence/http_allowed.pcapng

(root@1f1f5ef2d179)~/SBT-DF203-Lab6
#
```


9. Part C & D – Rule Implementation, Blocked Capture & Hit Counters

Inserting iptables -I INPUT 1 -s 127.0.0.1 -p tcp --dport 80 -j DROP resulted in immediate connection blocking:

- **Client Behavior:** curl received no response to its initial SYN packet and timed out after 5000ms (curl: (28) Connection timed out).
- **Packet Capture:** Showed repeated unacknowledged TCP SYN retransmissions from the client.
- **Rule Counters:** iptables -L -n -v recorded packet and byte increments on Rule #1, proving packet matching at the kernel Netfilter layer.



```
(root@ 1f1f5ef2d179)~/SBT-DF203-Lab6
# iptables -D INPUT -s 127.0.0.1 -p tcp --dport 80 -j DROP

(root@ 1f1f5ef2d179)~/SBT-DF203-Lab6
# iptables -C INPUT -s 127.0.0.1 -p tcp --dport 80 -j DROP && echo "Rule verified present." | tee reports/rule_verification.txt
Rule verified present.

(root@ 1f1f5ef2d179)~/SBT-DF203-Lab6
# tshark -i lo -f 'tcp port 80' -a duration:15 -w evidence/http_blocked.pcapng &
[1] 476

(root@ 1f1f5ef2d179)~/SBT-DF203-Lab6
# Running as user "root" and group "root". This could be dangerous.
Capturing on 'Loopback: lo'
sleep 2

[1]+ Done tshark -i lo -f 'tcp port 80' -a duration:15 -w evidence/http_blocked.pcapng

(root@ 1f1f5ef2d179)~/SBT-DF203-Lab6
# wait

(root@ 1f1f5ef2d179)~/SBT-DF203-Lab6
# iptables -L INPUT -n -v --line-numbers | tee reports/iptables_after_test.txt
Chain INPUT (policy ACCEPT 6 packets, 300 bytes)
num pkts bytes target prot opt in out source destination tcp dpt:80
1 0 0 DROP tcp -- * * 127.0.0.1 0.0.0.0/0 tcp dpt:80

(root@ 1f1f5ef2d179)~/SBT-DF203-Lab6
# sha256sum evidence/http_blocked.pcapng | tee reports/http_blocked_sha256.txt
59427b620b48565bdf06dacf1d11015c66204e69523cf974d257aab6d8f4be3 evidence/http_blocked.pcapng

(root@ 1f1f5ef2d179)~/SBT-DF203-Lab6
```

10. Part E & F – Comparison, Removal & Restoration

Indicator	Allowed Session (http_allowed.pcapng)	Blocked Session (http_blocked.pcapng)
Client SYN Sent	Yes	Yes
Server SYN-ACK Returned	Yes	No (Silently Dropped)
Handshake Status	Completed	Failed
HTTP Response	200 OK	None (Connection Timed Out)
Client Result	HTML Content Rendered	curl: (28) Timeout

Removing the rule via iptables -D INPUT -s 127.0.0.1 -p tcp --dport 80 -j DROP restored normal HTTP access, verified by a successful 200 OK curl response.

```
Terminal Shell Edit View Window Help
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sadique — root@1f1f5ef2d179: ~/SBT-DF203-Lab6 — docker exec -it sbtdf203-lab bash — 132x33

(root@1f1f5ef2d179)~/SBT-DF203-Lab6
# for PCAP in evidence/http_allowed.pcapng evidence/http_blocked.pcapng; do
> echo "==== $PCAP ====="
tshark -r "$PCAP" -Y 'tcp.flags.syn==1 || http.request || http.response || tcp.analysis.retransmission' \
-T fields -e frame.number -e frame.time_relative -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport \
-e tcp.flags -e tcp.analysis.retransmission -e http.request.uri -e http.response.code
done | tee reports/allowed_vs_blocked.tsv
==== evidence/http_allowed.pcapng ====
Running as user "root" and group "root". This could be dangerous.
==== evidence/http_blocked.pcapng ====
Running as user "root" and group "root". This could be dangerous.

(root@1f1f5ef2d179)~/SBT-DF203-Lab6
# iptables -D INPUT -s 127.0.0.1 -p tcp --dport 80 -j DROP

(root@1f1f5ef2d179)~/SBT-DF203-Lab6
# iptables -L INPUT -n -v --line-numbers | tee -a reports/iptables_restored.txt
Chain INPUT (policy ACCEPT 8 packets, 400 bytes)
num pkts bytes target prot opt in out source destination

(root@1f1f5ef2d179)~/SBT-DF203-Lab6
# curl -v --connect-timeout 5 http://127.0.0.1/firewall_lab.html 2>&1 | tee reports/restored_curl.txt
* Trying 127.0.0.1:80...
* Established connection to 127.0.0.1 (127.0.0.1 port 80) from 127.0.0.1 port 33444
% Total % Received % Xferd Average Speed Time Time Time Current
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 * using HTTP/1.x
> GET /firewall_lab.html HTTP/1.1
> Host: 127.0.0.1
> User-Agent: curl/8.21.0
> Accept: */*
>
```

```
Terminal Shell Edit View Window Help
MAY2026 MERGE Int. University of applied...ences.zip LETTER OF RECOM...RESS.pdf Parental Consent Form_M...ress.pdf cisco cyberanalytics.png cyberanalytics.pkt Research Process Jomacs - Version Control.pdf
sadique — root@1f1f5ef2d179: ~/SBT-DF203-Lab6 — docker exec -it sbtdf203-lab bash — 132x33

num pkts bytes target prot opt in out source destination

(root@1f1f5ef2d179)~/SBT-DF203-Lab6
# curl -v --connect-timeout 5 http://127.0.0.1/firewall_lab.html 2>&1 | tee reports/restored_curl.txt
* Trying 127.0.0.1:80...
* Established connection to 127.0.0.1 (127.0.0.1 port 80) from 127.0.0.1 port 33444
% Total % Received % Xferd Average Speed Time Time Time Current
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 * using HTTP/1.x
> GET /firewall_lab.html HTTP/1.1
> Host: 127.0.0.1
> User-Agent: curl/8.21.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Fri, 18 Sep 2026 04:31:52 GMT
< Server: Apache/2.4.68 (Debian)
< Last-Modified: Fri, 18 Sep 2026 02:25:15 GMT
< ETag: "73-65bb89c5b81b0"
< Accept-Ranges: bytes
< Content-Length: 115
< Vary: Accept-Encoding
< Content-Type: text/html
<
{ [115 bytes data]
100 115 100 115 0 0 622 0
* Connection #0 to host 127.0.0.1:80 left intact
<!DOCTYPE html>
<html><body><h1>ICDFA Network Forensics Firewall Lab</h1><p>Server: sbtdf203-lab</p></body></html>

(root@1f1f5ef2d179)~/SBT-DF203-Lab6
```

11. Forensic Interpretation & Discussion

1. **DROP vs. REJECT:** A DROP action silently discards matching packets without notifying the sender, causing TCP retransmissions and client timeouts. A REJECT action actively returns an explicit TCP RST or ICMP destination unreachable frame, causing immediate connection refusal (Connection refused).
2. **Rule Counter Value:** Rule counters corroborate that packet loss occurred due to an active firewall policy rather than network congestion or interface failure.
3. **Service Down vs. Firewall Blocked:** If Apache were stopped, client attempts would receive an immediate TCP RST from the operating system stack (Connection refused), whereas a DROP firewall rule results in silent timeout behavior.

12. Findings Worksheet

Question / Investigative Field	Forensic Finding
Applicable Chain	INPUT chain (traffic destined for the local host)
Applied Target	DROP
Observed Client Outcome	curl: (28) Connection timed out after 5003 ms
Observed Retransmission Pattern	Client sent repeated SYN frames without receiving SYN-ACK
Rule Counter Effect	Packet and byte counts incremented upon packet arrival
Restoration Verification	iptables -D executed; curl returned 200 OK

Academic Integrity and Authorization Statement

Student Declaration: I confirm that I completed this lab in an authorized environment, preserved the supplied evidence, did not target any third-party system, and accurately documented my own commands, observations, and conclusions.

Signed: Abubakari-Sadique Hamidu

Student ID: 2025-FWSD-11392

Date: 18 September 2026